

Depletion Width Formula

The depletion width (W) in a p-n junction can be expressed by the formula:

$$W = \sqrt{\frac{2\epsilon(V_0 + V)}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

Where:

- ϵ is the permittivity of the semiconductor material,
- V_0 is the built-in potential of the junction,
- V is the externally applied voltage (positive for forward bias and negative for reverse bias),
- q is the elementary charge (approximately 1.6×10^{-19} Coulombs),
- N_A and N_D are the acceptor and donor doping concentrations, respectively.